

Numerical Methods for Partial Differential Equations

A.Y. 2025/2026

Written exam - February 16, 2026

Maximum score: 26. Duration: 2h 15m.

Answers to questions must be provided on paper (handwritten). Source code and picture files should be uploaded to the dedicated WeBeep folder. **All uploaded files (including source code) must be appropriately referenced when answering to questions on paper** (e.g. “The solution is depicted in `exercise-1-plot.png`”).

Questions marked with (*) should be skipped by multichance students who have requested a 30 % reduction in the exam length.

Exercise 1.

Consider the problem

$$\begin{cases} -\operatorname{div}(\alpha \nabla u) - \boldsymbol{\beta} \cdot \nabla u + \gamma u = f & \text{in } \Omega, \\ \alpha \frac{\partial u}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega, \end{cases} \quad (1)$$

where $\alpha > 0$, $\boldsymbol{\beta} \in \mathbb{R}^2$ and $\gamma \geq 0$ are three given constant coefficients, while f is a given function and $\Omega \subset \mathbb{R}^2$.

1.1. [1 pt] Write the weak formulation of problem (1).

1.2. [3 pt] Prove under which conditions the weak problem obtained at point 1.1 has a unique solution.

1.3. [2 pt] (*) Approximate (1) by the finite element method, using polynomials of order 2. Discuss its stability, convergence and error estimates.

1.4. [1 pt] Write 1 iteration of the Dirichlet-Neumann method for problem (1) and comment on its convergence properties.

1.5. [1 pt] Let now $\Omega = (0, 1)^2$, and consider the following parameters:

$$\alpha = 1, \quad \boldsymbol{\beta} = (1, 1), \quad \gamma = 1.$$

Compute the forcing term f such that the exact solution to problem (1) is given by

$$u(x, y) = \cos(2\pi x) \cos(4\pi y)$$

1.6. [4 pt] Using `deal.II`, implement a finite element solver for problem (1). Use the data and function f from point 1.5. Use quadratic finite elements over the mesh that

can be found at this link: <https://github.com/michelebucelli/nmpde-labs-aa-24-25/blob/main/examples/gmsh/mesh-square-h0.100000.msh>. Upload the source code of the solver.

1.7. [2 pt] Using Paraview, generate and upload the following plots of the solution obtained at point 1.6:

1. a plot of the solution over the domain Ω ;
2. (*) a plot of the solution along the line $y = 0.25$ (**hint:** use the Paraview filter “Plot over line”).

1.8. [2 pt] Implement a function that computes the L^2 and H^1 norms of the error between the numerical and the exact solution. Upload the source code (it can be included in the source code uploaded for point 1.6, as long as it is clearly indicated).

1.9. [2 pt] (*) Using the function implemented at point 1.8 and the same data as in point 1.5, study the solver convergence by solving the problem on progressively more refined meshes. Use quadratic elements and the meshes found at the following link: <https://github.com/michelebucelli/nmpde-labs-aa-24-25/tree/main/examples/gmsh> (the meshes are named `mesh-square-hX.msh`, where X is the mesh size). Upload a plot of the error and comment on the results in relation to the theory. Include the comments as comments in the uploaded source code.

Exercise 2.

Consider the problem

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \mu \Delta \mathbf{u} + \nabla p + \mathbf{u} \cdot \nabla \mathbf{u} = \mathbf{0} & \text{in } \Omega, t \in (0, T), \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, t \in (0, T), \\ \mu \frac{\partial \mathbf{u}}{\partial \mathbf{n}} - p \mathbf{n} = \boldsymbol{\psi} & \text{on } \partial\Omega, t \in (0, T), \\ \mathbf{u} = \mathbf{0} & \text{in } \Omega, t = 0, \end{cases} \quad (2)$$

where $\boldsymbol{\psi}$ is a vector-valued function depending on both $\mathbf{x}$ and t , $\Omega \subset \mathbb{R}^2$, and $T > 0$.

2.1. [2 pt] Write the weak formulation of (2).

2.2. [1 pt] Write the spatial approximation of (2) using the following spaces:

$$\begin{aligned} V_h &= \{ \mathbf{v}_h \in [C^0(\Omega)]^2 : \mathbf{v}_h|_K \in [\mathbb{P}^2(K)]^2 \text{ for all } K \in \mathcal{T}_h \}, \\ Q_h &= \{ q_h \in C^0(\Omega) : q_h|_K \in \mathbb{P}^1(K) \text{ for all } K \in \mathcal{T}_h \}. \end{aligned}$$

2.3. [2 pt] Approximate the problem obtained at point 2.1 by the implicit backward Euler method in time for the linear terms and the explicit forward method in time for the non linear term $\mathbf{u} \cdot \nabla \mathbf{u}$. Is the problem obtained at every time-step well-posed? If so, why?

2.4. [3 pt] (*) Write the expected stability condition on Δt and error estimate for the problem obtained in point 2.3 in terms of both Δt and h .